

Module 24.13: Logistic Regression — Intuition & The Sigmoid Function

1. Why Linear Regression Fails for Classification

In binary classification problems, the target label is categorical: $y \in \{0, 1\}$. Attempting to fit a linear regression hypothesis $\hat{y} = \mathbf{w}^T \mathbf{x} + b$ directly creates two fundamental mathematical flaws:

Flaw 1: Unbounded Output Range

Linear regression outputs real values on $(-\infty, +\infty)$.

Probabilities must be strictly bounded in $[0, 1]$. A prediction of $\hat{y} = 2.4$

Flaw 2: Extreme Outlier Sensitivity

A single extreme positive outlier far to the right shifts the fitted regression tilting the decision boundary threshold and causing severe misclassification of

2. From Odds and Logits to the Sigmoid Function

To map continuous inputs $\mathbf{w}^T \mathbf{x} + b \in (-\infty, \infty)$ into valid calibrated probabilities $p \in (0, 1)$, we progress through three mathematical steps:

Step 1: Probability to Odds Ratio

The **Odds** of an event is the ratio of probability of occurrence to probability of non-occurrence:

$$\text{Odds} = \frac{p}{1-p} \in [0, \infty)$$

Step 2: The Logit Function (Log-Odds)

Taking the natural logarithm of the odds produces the **Logit**:

$$\text{Logit}(p) = \ln \left(\frac{p}{1-p} \right) \in (-\infty, \infty)$$

Step 3: Inverting the Logit (The Sigmoid Function)

We set the linear predictor equal to the log-odds:

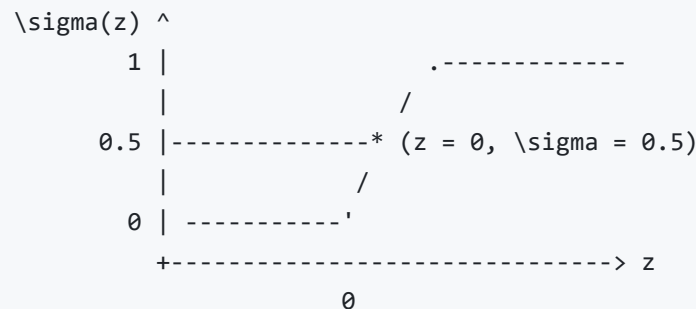
$$\ln \left(\frac{p}{1-p} \right) = \mathbf{w}^T \mathbf{x} + b = z$$

Exponentiating both sides:

$$\frac{p}{1-p} = e^z \implies p = e^z(1-p) \implies p(1+e^z) = e^z$$

Solving for probability p :

$$p = \sigma(z) = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}$$



3. Mathematical Properties of the Sigmoid Function

1. **Symmetric S-Curve:** Centered at the inflection point $(0, 0.5)$.
2. **Asymptotes:** $\lim_{z \rightarrow \infty} \sigma(z) = 1$ and $\lim_{z \rightarrow -\infty} \sigma(z) = 0$.
3. **Analytic Derivative:** The derivative expresses cleanly in terms of the function itself:

$$\sigma'(z) = \frac{d}{dz} \left(\frac{1}{1+e^{-z}} \right) = \frac{e^{-z}}{(1+e^{-z})^2} = \sigma(z)(1-\sigma(z))$$

4. The Linear Decision Boundary

To assign discrete binary class labels $\hat{y} \in \{0, 1\}$, we apply a standard probability threshold $\tau = 0.5$:

$$\hat{y} = \begin{cases} 1 & \text{if } P(y = 1|\mathbf{x}) \geq 0.5 \\ 0 & \text{if } P(y = 1|\mathbf{x}) < 0.5 \end{cases}$$

Because $\sigma(z) \geq 0.5 \iff z \geq 0$, the decision boundary satisfies:

$$\mathbf{w}^T \mathbf{x} + b = 0$$

This equation defines a **hyperplane** partitioning feature space into two prediction regions.